

Defocus-Aware Modeling and Control Analysis of a QD-Based Optical Tracking System: Experimental and Simulated Evaluation Using the DOLCE Terminal

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Abstract—Free-space optical communication (FSOC) is attracting significant attention as a core technology for future space communication infrastructure, owing to its capability of providing wide bandwidth and high antenna gain with narrow beam divergence. However, the small divergence angle of optical beams makes the link highly sensitive to pointing errors, requiring highly accurate tracking sensors. The quadrant detector (QD), with its simple structure and fast response, is a strong candidate for fine-tracking, but its linear response range is severely limited when placed at the focal plane. To address this limitation, this study investigates a QD defocus method in which the QD is intentionally placed out of focus to expand its field of view. Using wave-optics simulations and laboratory experiments, the response characteristics of QDs under defocus

conditions are systematically evaluated. Furthermore, acquisition and tracking experiments were conducted using a numerical simulator based on the DOLCE optical terminal configuration to verify the validity of QD defocus. The results clarify the trade-off between expanded linear range and tracking accuracy, providing valuable insights for the design of compact and cost-effective optical communication terminals.

Index Terms—Free-space optical communication (FSOC), Pointing, Acquisition, and Tracking (PAT), Quadrant Detector (QD), Defocus, Terminal Simulator

I. BACKGROUND

In recent years, the demand for high-capacity data transmission between satellites and between satellites and the ground

has been rapidly increasing, driving active research and development in free-space optical communication (FSOC) [1] [2]. Compared with radio frequency (RF) communication, optical communication offers a wider bandwidth and higher antenna gain due to its shorter wavelength [1], making it a promising core technology for next-generation space communication infrastructure. However, the extremely narrow beam divergence of optical links makes them highly sensitive to pointing errors and relative motion between terminals, necessitating the implementation of highly accurate pointing, acquisition, and tracking (PAT) functions [1].

For fine-tracking in PAT systems, sensors such as camera-based imagers and position-sensitive detectors (PSDs) have been widely used [3]. Among these, the quadrant detector (QD) has attracted particular attention as a fine-tracking sensor for spaceborne optical communication terminals because of its simple structure and fast response [4]. A QD consists of four photodiodes arranged in a quadrant configuration, and the incident beam position can be estimated from the differential ratios of photocurrents from each quadrant. This enables rapid acquisition of angular error signals, which is advantageous for compensating attitude fluctuations and mechanical vibrations.

One of the major limitations of the QD, however, is its extremely narrow linear response range. When placed at the focal plane of the front lens, the region in which the output signal maintains an approximately proportional relationship is limited by the diffraction of the optics and typically extends only several tens of microradians. Consequently, when performing fine-tracking under large attitude errors of the platform, the effective operating range becomes severely restricted.

To address this limitation, this study focuses on a QD defocus method, in which the QD is intentionally placed out of the focal plane. In this configuration, the spot size increases and the linear response range has the potential to expand, while the intensity distribution deviates from the simple flat-top or Airy-disk profile and becomes nonlinear. Most previous studies have relied on idealized distribution models, and detailed evaluations of QD response characteristics under realistic defocus conditions remain limited. Therefore, accurately characterizing QD behavior under defocus is essential to provide practical insights toward expanding the effective field of view (FOV) of the fine-tracking sensor while maintaining high accuracy.

The objective of this study is to systematically evaluate the sensitivity characteristics and linear response range of the QD under defocus conditions through theoretical modeling, numerical simulations, and laboratory experiments. In the simulations, a wave-optics-based propagation model is employed to generate spot intensity distributions for various defocus distances, which are then used to calculate QD voltage outputs and construct a nonlinear response model. In the experiments, a QD is mounted on an optical bench, and its relative distance from the focal plane is varied while injecting an optical beam to directly measure response variations with defocus. These experiments validate the simulation model and quantitatively assess the extent of FOV expansion under defocus.

Furthermore, acquisition and tracking analyses are con-

ducted using a numerical simulator that reproduces the optical configuration of the DOLCE terminal [5], a ground-based optical communication testbed developed at the University of Tokyo. The DOLCE terminal incorporates a fine-pointing mirror and dynamic disturbance models, enabling realistic reproduction of integrated PAT control dynamics.

In future work, tracking control experiments using the actual DOLCE terminal will be conducted to verify the effectiveness of the QD defocus approach under real operating conditions. As a preliminary step, this paper focuses on the QD-level defocus response experiments and numerical simulations to deepen the understanding of QD sensor behavior and provide fundamental insights for future optical terminal design guidelines.

In summary, this study comprehensively analyzes the defocus behavior of QDs from both optical and sensor-response perspectives, thereby providing fundamental knowledge toward achieving a balance between wide FOV and high-precision tracking performance in optical communication terminal design.

II. PRINCIPLE OF QD

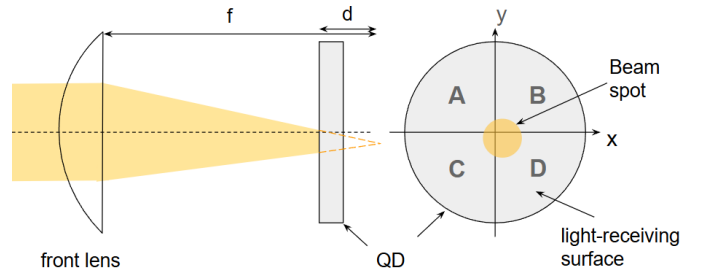


Fig. 1: Schematic of a QD-based fine-tracking sensor

A QD consists of four photodiodes arranged in a quadrant configuration. The photocurrents generated in each segment are converted into voltages by operational amplifiers, allowing the incident beam position to be detected. Compared to other optical sensors, the QD is characterized by its high angular resolution and fast response speed. Figure 1 illustrates a typical configuration when a QD is employed as a fine-tracking sensor. Here, the focal length of the front lens is denoted as f , and the defocus, defined as the distance from the focal plane to the detector surface, is denoted as d .

As shown in the figure, the incident beam is focused by the front lens and directed onto the QD, which is placed near the focal plane. When the incident beam deviates in angle, the spot position on the QD shifts accordingly. This displacement changes the ratio of the optical power received by the four quadrants, from which the beam spot position can be estimated. When the spot is located near the center of the QD, the output signals of the quadrants are approximately proportional to the displacement of the spot from the center. If we denote the output voltages corresponding to the four

quadrants A–D as v_a-v_d , the beam spot position (p_x, p_y) can be expressed as follows [6]:

$$p_x = P_0 \frac{(v_a + v_d) - (v_b + v_c)}{v_a + v_b + v_c + v_d}, \quad (1)$$

$$p_y = P_0 \frac{(v_a + v_b) - (v_c + v_d)}{v_a + v_b + v_c + v_d} \quad (2)$$

Here, P_0 represents the region in which the QD provides a linear response as a tracking sensor, referred to as the linear FOV. When the spot on the QD is assumed to be an idealized circular flat-top spot with diameter d_s , simple geometric analysis shows that the proportionality constant is given by $K_{flat} = \pi/8 \simeq 0.393$, i.e.,

$$P_0 = K_{flat} \cdot d_s = \frac{\pi}{8} d_s \quad (3)$$

In satellite optical communication, the incident beam can be regarded as a collimated top-hat beam. When the QD is placed exactly at the focal plane of the lens under this conditions, the tracking accuracy is ultimately limited by the diffraction condition. The focused spot then forms an Airy distribution, and using the Airy disk diameter (first dark ring), defined as $d_A = 1.22\lambda f/R$ (where R is the aperture radius and λ is the wavelength), P_0 can be expressed as [7]:

$$P_0 = \frac{3\pi}{32} \cdot \frac{\lambda f}{R} = \frac{1}{4.14} d_A \quad (4)$$

In practice, however, when the QD is placed away from the focal plane, the spot does not become flat but exhibits a non-uniform intensity distribution. Thus, the response does not fully agree with either of the above expressions. To address this, we define a conversion coefficient K_T under defocused conditions, and P_0 is expressed in terms of the geometric spot diameter d_s as:

$$P_0 = K_T \cdot d_s \quad (5)$$

On the other hand, in short-range ground-based optical communication experiments, or when a general laser beam is directly incident onto the system, the incident light can be regarded as a Gaussian beam. In this case, the spot formed on the QD after passing through the front lens also exhibits a Gaussian intensity distribution, whose size varies depending on the defocus amount. Although several models have been proposed to describe the QD response for Gaussian beams [8], from the standpoint of practical hardware implementation and model simplicity, it can be expressed in a form similar to the above equations as:

$$P_0 = K_G \cdot d_w \quad (6)$$

Here, K_G denotes the conversion coefficient for the Gaussian beam, and d_w represents the beam diameter at which the radiant intensity of the Gaussian profile on the QD surface falls to $1/e^2$ of its peak value.

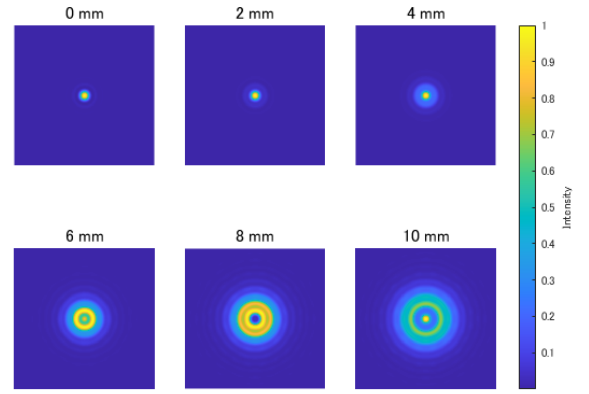
Based on the estimated spot positions, the corresponding beam directions (q_x, q_y) are obtained by dividing them by the focal length of the lens.

$$q_x = \frac{p_x}{f}, \quad q_y = \frac{p_y}{f} \quad (7)$$

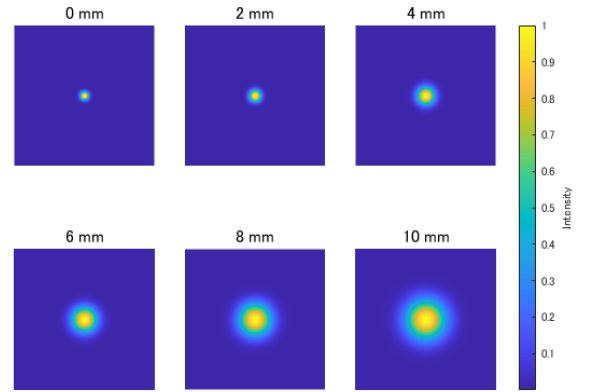
III. QD NUMERICAL SIMULATOR

A. Analysis Method

This section describes a numerical simulator developed to model the output characteristics of the QD under various spot conditions. The simulator calculates the relative output voltages of each quadrant cell by integrating the optical intensity incident on each cell when the spot image is placed at a given position on the QD surface. By applying the position estimation algorithm described in the previous section to these simulated outputs, the behavior of the linear FOV as well as the estimated spot position can be analyzed. The optical parameters and incident beam intensities are set to be identical to those used in the experiment (see Table III).



(a) For the top-hat incident beam.



(b) For the Gaussian incident beam.

Fig. 2: Simulated spot intensity distributions under different defocus conditions. Each image corresponds to a FOV of $0.8 \text{ mm} \times 0.8 \text{ mm}$.

In accordance with the architecture of the DOLCE terminal, two types of incident beams are considered. For satellite communication scenarios, the incoming light is modeled as a top-hat beam with a 3 mm diameter, corresponding to the telescope aperture. For ground-based short-range experiments, the beam is modeled as a Gaussian beam with a beam waist

diameter of 2.27 mm, consistent with the fiber collimator used in the actual setup.

In the simulation experiment, the first step is to generate the intensity distributions of the spot on the QD for different defocus conditions, for both top-hat and Gaussian beams, based on wave optics propagation calculations. In the case where the incident beam is top-hat beam, the spot follows an Airy distribution at zero defocus, while under defocused conditions, it exhibits a non-flat intensity profile rather than a uniform flat-top shape. In the case where the incident beam is Gaussian, the resulting spot maintains a Gaussian intensity distribution, as clearly confirmed in the simulation results.

Next, for each spot distribution, the simulator computes the cell output voltages of quadrants A–D as the spot is shifted along the QD sensor's X-axis, as illustrated in Fig. 3. The position estimation algorithm is then applied to the calculated outputs. This procedure enables the evaluation of both the estimation accuracy and the extent of the linear FOV.

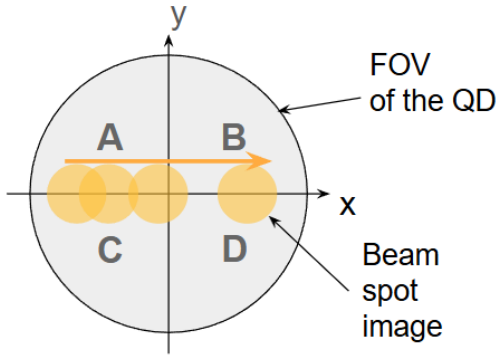


Fig. 3: Illustration of spot displacement across the QD sensor in the simulation.

B. Simulation Results

This section presents representative outputs from the QD simulator and discusses the estimation of the optimal conversion coefficients K_T and K_G under defocused conditions. The aim is to achieve position estimation with higher accuracy than that provided by the conventional flat-spot assumption.

1) Case of Top-hat Incident Beam:

Figure 4 presents examples of the QD output voltage ratios as a function of the spot center position for several defocus conditions. As the defocus increases, the spot size becomes larger, resulting in a shallower slope in the response curves and an expanded linear FOV. At the same time, changes in the spot intensity distribution induced by defocus manifest as additional inflection points in the response curves.

Next, the conversion from voltage ratio to estimated position is examined. Figure 5 shows the estimated spot positions p_x and estimation errors ($p_x - x_{true}$) when applying the conversion defined by Eq. 4 to the voltage ratio at $d = 0$. For $d \neq 0$, the estimated positions and errors obtained with conversion coefficients $K_T = 0.3, 0.35, \pi/8 (\simeq 0.392)$ are shown in Fig. 6.

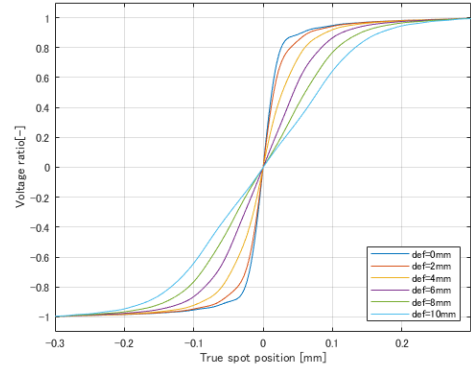
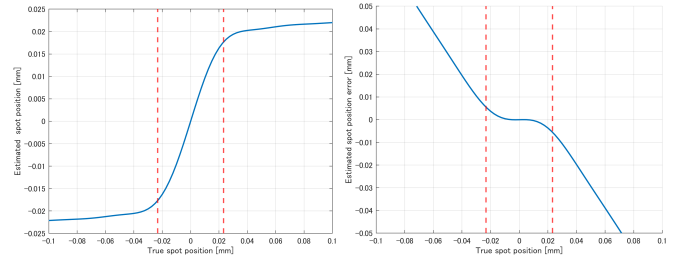


Fig. 4: QD output voltage ratios under various defocus conditions for the top-hat incident beam.



(a) Estimated position

(b) Estimation error

Fig. 5: Simulation results at $d = 0$ mm for top-hat incident beam. (the red dashed lines indicate the linear FOV range).

For the Airy distribution case ($d = 0$), Eq. 4 provides an excellent match to the simulated response. Under defocused conditions, however, the optimal coefficient depends on the degree of defocus: for $d = 2$ mm, where the spot shape remains close to an Airy distribution, $K_T = 0.35$ provided the best agreement, while for $d = 10$ mm, where the spot shape resembles a flat-top distribution, $K_T = \pi/8$ matched more closely. These results demonstrate that the optimal conversion coefficient varies with the amount of defocus.

Table I summarizes the conversion coefficients that minimize the root-mean-square error (RMSE) of the estimated position within the linear FOV for several defocus values. The results confirm that larger defocus leads to larger optimal K_T . Although the linear FOV increases with defocus, the RMSE within the linear FOV also increases, revealing a trade-off between linear FOV and estimation accuracy. While the table lists the coefficients that minimize RMSE, in practice, if a tolerance for estimation error is specified, one can instead select K_T to maximize the linear FOV while still satisfying the error requirement.

2) Case of Gaussian Incident Beam:

Similarly, Fig. 7 shows the QD output voltage ratios for several defocus conditions when the incident beam is Gaussian. As in the case of the top-hat beam, increasing the defocus results in a larger spot size and a gentler slope in the response curves. However, because the intensity gradient

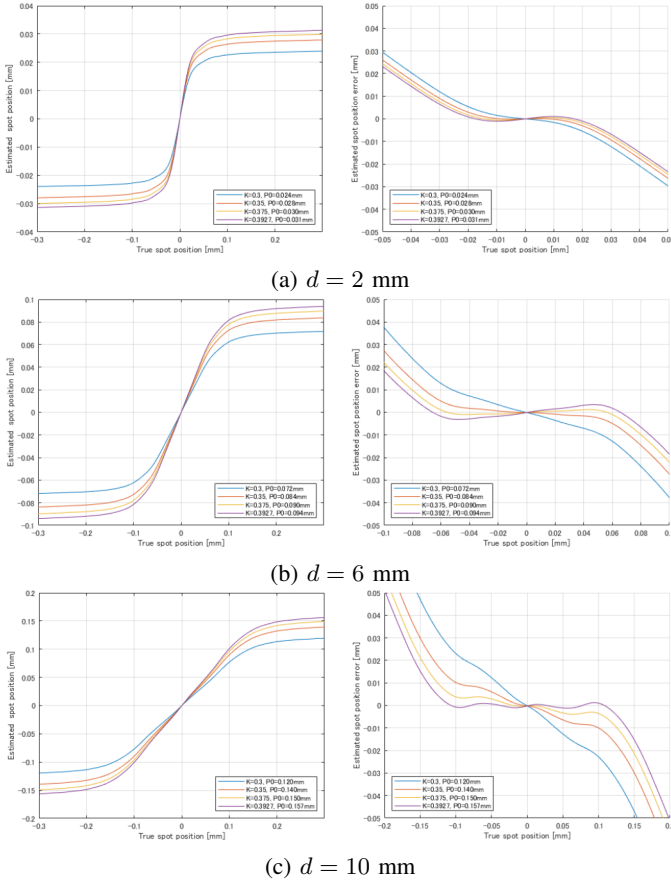


Fig. 6: Estimated spot positions (left) and errors (right) under defocused conditions with various conversion coefficients K_T .

TABLE I: Summary of optimal conversion coefficients K_T , linear FOV, and RMSE for different defocus values.

Defocus [mm]	2	4	6	8	10
d_s [mm]	0.08	0.16	0.24	0.32	0.40
Optimal K_T	0.343	0.390	0.395	0.398	0.406
Linear FOV [mm]	0.027	0.062	0.095	0.13	0.16
RMSE [10^{-3} mm]	2.71	4.50	4.56	4.62	4.82

of the Gaussian beam is inherently smoother than that of a top-hat beam, the resulting curves exhibit fewer inflection points and appear more continuous and gradual.

For the Gaussian beam, the optimal conversion coefficient K_G that minimizes the RMSE of position within the linear FOV was also evaluated for several defocus conditions, and the results are summarized in Table II. Unlike the case of the top-hat beam, the optimal K_G remains nearly constant regardless of the amount of defocus. This behavior is attributed to the fact that, in contrast to the top-hat case, the Gaussian intensity profiles at different defocus values are nearly self-similar. The trade-off between the linear FOV and estimation accuracy, however, follows the same trend as that observed for the top-hat beam.

As an example, Fig. 8 shows the estimated spot positions and corresponding estimation errors obtained using the optimal

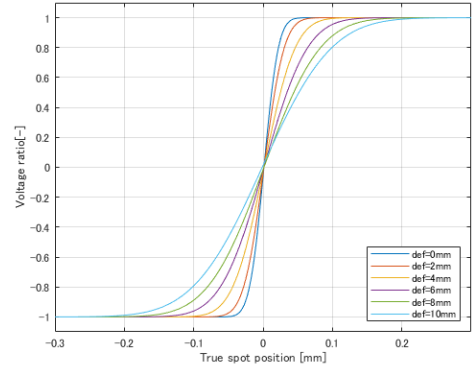


Fig. 7: QD output voltage ratios under various defocus conditions for Gaussian incident beam.

TABLE II: Summary of optimal conversion coefficients K_G , linear FOV, and RMSE for different defocus values.

Defocus [mm]	0	2	4	6	8	10
d_w [mm]	0.033	0.046	0.071	0.098	0.127	0.157
Optimal K_G	0.344	0.354	0.354	0.355	0.353	0.358
Linear FOV [mm]	0.022	0.032	0.050	0.070	0.090	0.112
RMSE [10^{-3} mm]	1.03	1.46	2.38	3.43	4.53	5.55

K_G for several defocus conditions. The results confirm that the estimation errors are effectively suppressed and remain stable across different defocus values.

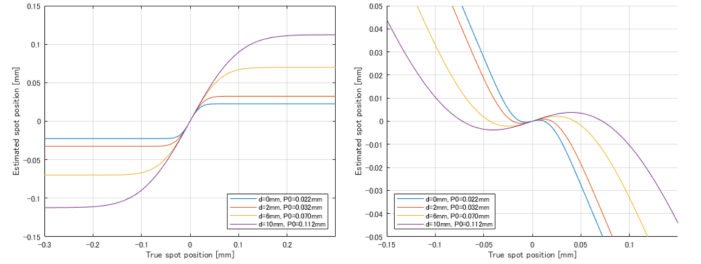


Fig. 8: Estimated spot positions (left) and estimation errors (right) with the optimal K_G under different defocus conditions.

IV. QD EXPERIMENT

A. Experimental Configuration

To validate the results obtained from the simulations in Section III, an experiment using an actual QD was conducted. The experimental configuration is shown in Fig. 9. In the setup, a 1550 nm laser beam with a 2.27 mm beam diameter emitted from a fiber collimator was focused by a front lens and then incident onto the QD, which was mounted on a three-axis stage positioned near the focal plane.

First, with the defocus position fixed, the QD was translated horizontally in steps of 0.01 mm, similar to the procedure used in the simulation, and its output voltage corresponding to the spot position was read out and recorded by Arduino. The defocus distance was varied by moving the front lens, which was mounted on a micro-stage, back and forth along

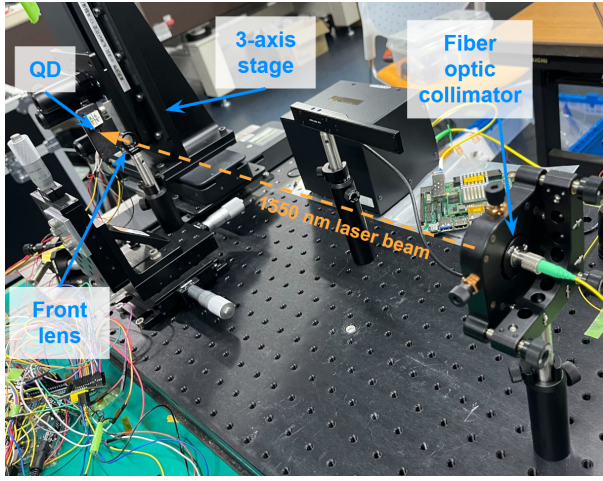


Fig. 9: Experimental configuration of the QD setup.

the optical axis. Before moving the QD horizontally, the QD position is adjusted so that the spot is centered on the QD, while monitoring the raw output values from the QD. The change in defocus distance is achieved by moving the front lens mounted on the micro-stage back and forth along the optical axis.

The specifications of the QD (KPDE150Q-H15) and the inverting amplifier used in the experiment are summarized in Table III.

TABLE III: Specifications of the QD and inverting amplifier.

Quadrant Detector (QD)	
Active area diameter	1.5 mm
Element gap	0.10 or 0.04 mm
Responsivity (@1550 nm)	1.0 A/W
I/V conversion resistance	200 k Ω
Dark current	0.3 nA
Inverting Amplifier	
R_1	22 k Ω
R_2	39 k Ω
C	0.0001 μ F

The focal length of the front lens for the QD was set to 75 mm, consistent with the simulation described later. The laser power was adjusted to 18.6 dBm using an optical attenuator, such that the QD would just reach saturation when all the laser power was incident on a single quadrant.

The entire experimental apparatus was placed on a vibration-isolated optical table. During the measurements, the surrounding environment was darkened by switching off nearby electrical lighting and covering windows with blackout curtains to minimize background light.

B. Experimental Results

The response characteristics of the QD at each defocus position were evaluated by plotting the horizontal coordinate of the QD, driven laterally by a three-axis stage, on the horizontal axis, and the horizontal voltage ratio on the QD (defined as p_x/P_0 in Eq. 1) on the vertical axis. At each coordinate point, measurements were repeated ten times with

the stage held stationary, and the average value was used. Because the translational resolution of the stage is limited to 1 μ m, the incident spot cannot always be perfectly symmetric between the upper and lower quadrants of the QD. Therefore, the obtained voltage-ratio curves were horizontally offset such that they pass through the origin for normalization.

As a preliminary experiment to determine the focal position, the distance between the QD sensor and the front lens was varied, and the corresponding output responses were measured. The micrometer position of the front lens was adjusted in 0.01 mm increments around the focal plane, and a subset of the obtained QD responses is shown in Fig. 10. The legend indicates the micrometer readings of the front-lens stage (e.g., “1200” corresponds to 12.00 mm). From the figure, it is observed that the slope within the linear region is maximized at a micrometer position of 12.00 mm, and that the response curves at 9.00 mm and 15.00 mm nearly coincide. Therefore, this position was defined as the focal position (defocus = 0 mm).

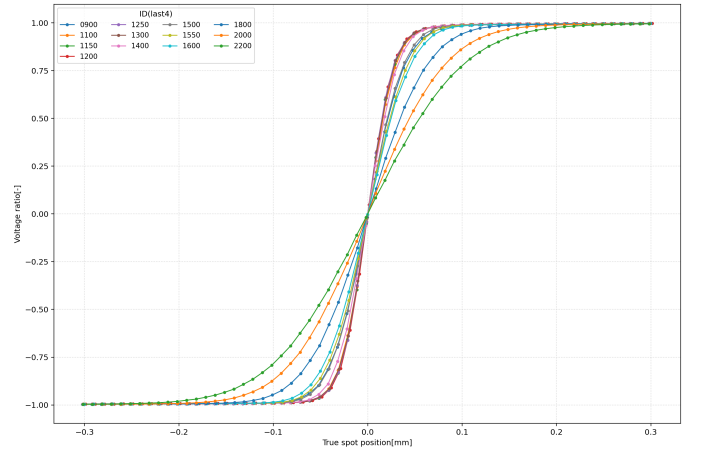


Fig. 10: Measured QD response curves for different front-lens positions used to identify the focal plane.

Next, the QD voltage responses were measured at defocus distances of 2, 4, 6, 8, and 10 mm relative to the defined focal position. The results are shown in Fig. 11. In each graph, the measured data (orange) are plotted together with the simulated results based on the proposed Gaussian-beam model (blue) and, for reference, the flat-spot model (green). Note that at the focal position (defocus = 0 mm), the geometrical spot diameter of the flat-spot model becomes zero, and therefore its curve is omitted.

From Fig. 11, it can be observed that the proposed Gaussian-beam model agrees well with the measured results across all defocus conditions, while the flat-spot model shows significant deviation, particularly in the nonlinear region. To quantitatively evaluate the agreement, the RMSE between the experimental data and each model was calculated within the linear FOV, defined as $P_0 = \pi/8 \times d_s$, where d_s is the spot diameter. The results are summarized in Table IV. The RMSE values of the proposed Gaussian-beam model are smaller than those of the flat-spot model for all defocus distances,

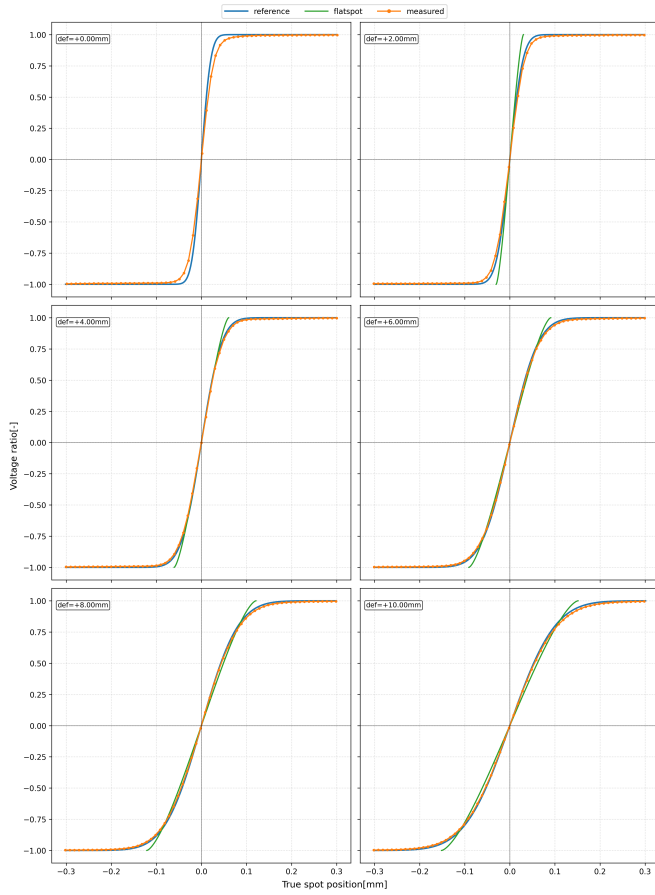


Fig. 11: Measured and simulated QD voltage-ratio responses for different defocus conditions. The measured data (orange) are compared with the Gaussian-beam model (blue) and the flat-spot model (green).

demonstrating the validity of the proposed model. (Since the plotted quantity is a voltage ratio, the RMSE is dimensionless.)

TABLE IV: RMSE of measured and simulated QD voltage-ratio responses at each defocus distance.

Defocus [mm]	0	2	4	6	8	10
RMSE of gaussian model	-	0.047	0.018	0.014	0.014	0.014
RMSE of flat spot	-	0.158	0.027	0.026	0.030	0.034

C. Discussion

In the standalone QD simulations, an increase in defocus resulted in an expansion of the linear response range while simultaneously decreasing sensitivity and increasing position estimation error. This behavior demonstrates the inherent trade-off caused by spot broadening when the detector is placed out of focus. To achieve both a wide field of view (FOV) and sufficient detection accuracy, appropriate selection of the defocus distance and conversion coefficients (K_T , K_G) must be made according to system requirements.

When comparing the simulated QD responses with experimental measurements, the proposed Gaussian beam model

showed excellent agreement under all defocus conditions, including in nonlinear regions where the conventional flat-spot model exhibited significant deviations. The improved agreement confirms that the actual spot intensity distribution is dominated by a Gaussian-type profile rather than a uniform flat-top distribution. Quantitatively, the root-mean-square (RMS) error within the linear FOV was consistently smaller for the Gaussian model, validating its effectiveness for both optical and control analyses.

At the focal position (defocus = 0 mm), the measured response exhibited a slight slope deviation from the simulation. This discrepancy is primarily attributed to the focus-determination procedure. In this experiment, the focal position was not determined with fine 0.1-mm steps; instead, a reference position x_0 was chosen where the response curves at ± 3 mm defocus appeared visually symmetric. Consequently, the actual optical focus might have been slightly offset. Additionally, imperfections in the front-lens surface shape or minor optical-axis misalignment during setup may have prevented ideal spherical focusing, further contributing to the observed deviation.

V. PAT NUMERICAL SIMULATION

A. Numerical Simulator Overview

Finally, the results of incorporating the developed QD model in Section III into the acquisition-and-tracking simulator, referred to as the DOLCE simulator, are presented. The DOLCE simulator is a numerical acquisition-and-tracking simulator developed in *MATLAB Simulink*. The system models a Pointing, Acquisition, and Tracking (PAT) configuration as illustrated in Fig. 12, where the QD sensor serves as the fine-tracking element. In addition to the QD, the simulator incorporates optical components such as the Fine Pointing Mirror (FPM), the telescope optics, and attitude disturbances representing spacecraft dynamics. This enables realistic reproduction of both optical and mechanical disturbances encountered in actual satellite systems.

Since the present simulator is currently intended for the ground-based experimental setup of the DOLCE terminal, it is configured to emulate short-range, face-to-face optical link tests on an optical bench. In such short-range configurations, the beam incident on the QD exhibits a Gaussian intensity distribution, and therefore the QD sensor model adopted here corresponds to the Gaussian-beam case proposed in Section III-B2. The model parameters are set to match the actual optical configuration of the DOLCE system: a defocus distance of 3.5 mm and an optimal conversion coefficient of $K = 0.351$. In future outdoor experiments using DOLCE, where long-range optical communication will be conducted, the QD model will be switched to the top-hat beam case to represent a collimated optical link.

The control architecture is based on a feedback loop that drives the FPM according to the centroid position detected by the QD sensor. A proportional–integral–derivative (PID) controller is used, and its parameters were designed considering the dynamic characteristics of the FPM to ensure sufficient

gain and phase margins. Since this simulator is currently tailored for the ground-based DOLCE experimental setup, the control and optical parameters are configured accordingly.

The control architecture is based on a feedback loop that drives the FPM according to the centroid position detected by the QD sensor. A proportional–integral–derivative (PID) controller is used, and its parameters were designed considering the dynamic characteristics of the FPM to ensure sufficient gain and phase margins. Its transfer function $C(s)$ is given by

$$C(s) = K_p \left(1 + \frac{s}{\omega_d}\right) \left(1 + \frac{\omega_i}{s}\right) \quad (8)$$

To set the controller gains, frequency-domain analysis was conducted with the aim of maximizing the control performance, given the stringent requirements of fine tracking. As a result, the following gain parameters were adopted:

TABLE V: Gain parameters of the fine-tracking controller

Parameter	Symbol	Value
Proportional gain	K_p	1000 V/rad
Derivative corner frequency	ω_d	300 Hz
Integral corner frequency	ω_i	1000 Hz

Since this simulator is currently tailored for the ground-based DOLCE experimental setup, the control and optical parameters are configured accordingly.

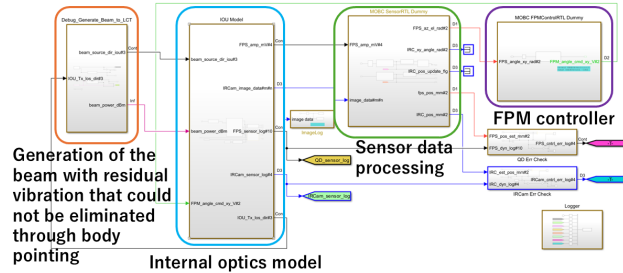


Fig. 12: Block diagram of the DOLCE PAT simulator incorporating the QD sensor model, FPM, and disturbance inputs.

B. Results

The total simulation time was 0.3 s, with the control loop activated at $t = 0.02$ s. Figure 13 shows the time history of the overall tracking error, defined as the root-sum-square (RSS) of the x - and y -axis angular errors. Before control activation ($t < 0.02$ s), the effect of simulated attitude and optical disturbances appears directly as tracking error. After control engagement, the error rapidly decreases owing to the feedback action of the PID controller. The steady-state tracking error achieved after stabilization was $14 \mu\text{rad}$, which satisfies the performance requirement assumed for the DOLCE ground experiment.

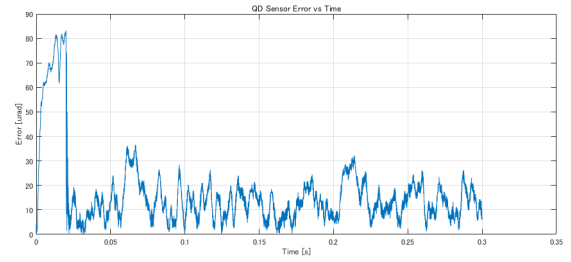


Fig. 13: Time history of the composite tracking error in the DOLCE PAT simulation. Control activation occurs at $t = 0.02$ s, after which the error converges to $14 \mu\text{rad}$.

C. Discussion

In the acquisition-tracking control analysis using the DOLCE simulator, stable and accurate performance was achieved when the defocus distance was set to 3.5 mm and the conversion coefficient to $K = 0.351$. After the control loop was engaged at $t = 0.02$ s, the tracking error rapidly converged, reaching a steady-state composite error of $14 \mu\text{rad}$. This result demonstrates that the accurate reproduction of QD response through the Gaussian beam model directly contributes to high-precision control design.

When the same control parameters derived from the flat-spot model were applied to the Gaussian-beam case, the control loop diverged. This indicates that neglecting the nonlinearity and sensitivity variation of the defocused QD can lead to instability in practical systems. Therefore, accurate modeling of the QD response—including its dependence on defocus and intensity distribution—is essential for designing robust and stable fine-tracking control systems.

Overall, the findings highlight that QD behavior under defocus involves optical wave-propagation effects that cannot be captured by simple geometric optics. The proposed model quantitatively represents these effects and thus provides a practical foundation for sensor and control design in compact optical terminals.

VI. CONCLUSION

This study investigated a QD defocus method for extending the linear response range of a QD and analyzed its implications for fine-tracking control in optical communication terminals. The main findings are summarized as follows:

- 1) **Model Development and Validation:** Wave-optics-based models for both top-hat and Gaussian beams were established to reproduce QD voltage-ratio outputs under various defocus conditions. Comparison with experimental measurements confirmed that the Gaussian-beam model closely matches the actual QD response, outperforming the conventional flat-spot approximation.
- 2) **Quantification of Defocus Effects:** The increase in defocus distance was shown to expand the linear response range while reducing detection sensitivity. This trade-off between linear FOV and accuracy was quantitatively characterized through RMS error evaluation.

- 3) **Verification Through Tracking Simulation:** Numerical acquisition-and-tracking experiments using the DOLCE simulator verified that stable control is achieved under the condition of defocus = 3.5 mm and $K = 0.351$. The resulting tracking error of 14 μrad satisfies the performance requirement for ground-based optical experiments.
- 4) **Design Implications:** The defocus method enables an expanded sensor field of view without the need for complex multi-sensor architectures. It thus offers a promising approach for simplifying and miniaturizing future optical communication terminals while maintaining sufficient tracking accuracy.

Future work will integrate the proposed QD response model into an actual optical terminal to validate tracking control performance and environmental robustness in real systems. In particular, evaluating stability under external disturbances such as satellite attitude fluctuations will further clarify the applicability and limitations of the defocus technique for spaceborne optical links.

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